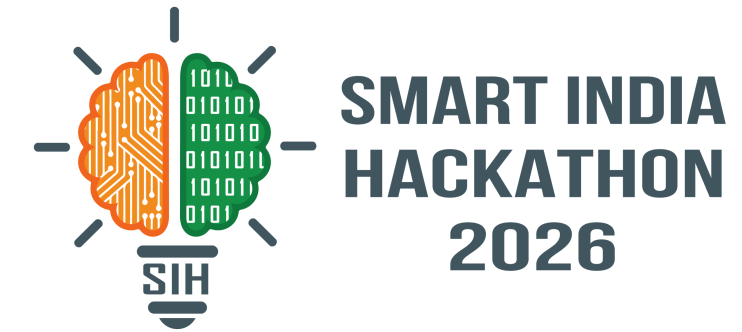


SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – 26151

Problem Statement Title- Dark web threat actor de-anonymization

Theme- Blockchain and Cybersecurity

PS Category- Software

Team ID-

Team Name- DeCypher



PROPOSED SOLUTION



Continuous Footprint Collection

Automated collection of dark-web activity, identifiers & infrastructure indicators with source tagging and scan tracking.



Cross Platform Actor Correlation

Neo4j graph links handles, PGP keys, wallets & marketplaces to reveal hidden connections and organized clusters.



AI Powered Persona Attribution

Stylometric & behavioral analysis to attribute personas, compute match likelihood and apply contradiction checks.



Infra & Real World Attribution

Correlates infra fingerprints (SSL certs, status pages, banners) to real-world assets and generates evidence-backed leads.



Explainable & Prioritized Intelligence

Confidence-scored, evidence-backed insights with timelines and evolution to preserve source provenance.



Autonomous, Self-Prioritizing Operation

Continuous re-scans, prioritization of high-risk actors and investigator feedback loop.



Investigator Dashboard

Interactive graph & timeline, filters, profiles and exportable reports (CSV/JSON/PDF).



Security & Governance by Design

Access-controlled, audit-logged intelligence built only on ethical academic & synthetic datasets. No live illegal-market scraping.

NEED FOR SOLUTION

The dark web has become a preferred operating space for threat actors in the modern age, mainly because it lets them hide their identity behind Tor hidden services, which makes attribution of threat actors operating on darkweb the main challenge for any investigation.



Drugs &
arm sales



Stolen data &
Hacking
Services



Money
Laundering



Terror
Financing



and other
illicit
activities



THE OBJECTIVE OF OUR SOLUTION IS TO:

Build a system for the **deanonymization of dark web threat actors** and link them to suspect **real-world entities** by correlating multi-source data, attributing personas and infrastructure, and delivering **explainable, prioritized intelligence** for investigations.

USP



Priority Score

Combines attribution confidence with threat severity, so investigators know what to act on first.



Actor Timeline

Reconstructs an actor's lifecycle from first seen to latest activity with handle evolution.



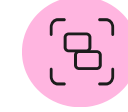
Evidence Trail

Every match shows which signals contributed and how strongly (shared wallet, stylometry, similarity %).



Contradiction/De-confliction Detection

Actively checks for conflicting evidence to reduce false positives.



Organized-cluster Detection

Detects and visualizes entire linked groups, not just pairwise matches.



Self-prioritizing Autonomous Mode

System decides what gets rescanned and prioritized based on confidence and change signals.



Investigator feedback loop

Human-in-the-loop confirmations/rejections reweight future scoring.



Access Control + Audit Logging

Role-based access with immutable audit logs for sensitive intelligence.



Ethical-by-Design

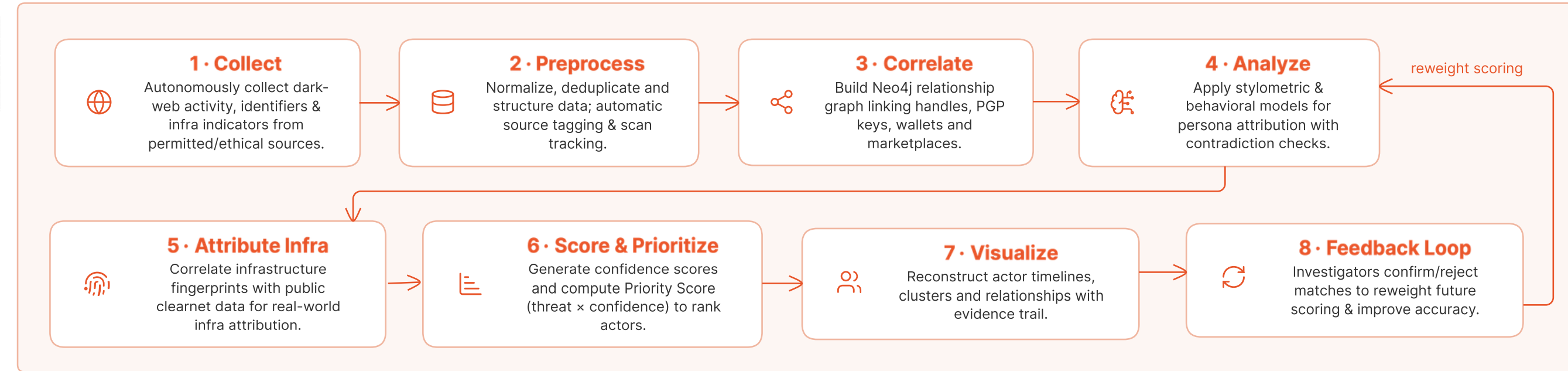
Built and demoed on academic/synthetic datasets within human-in-the-loop framing.

TECHNOLOGIES USED

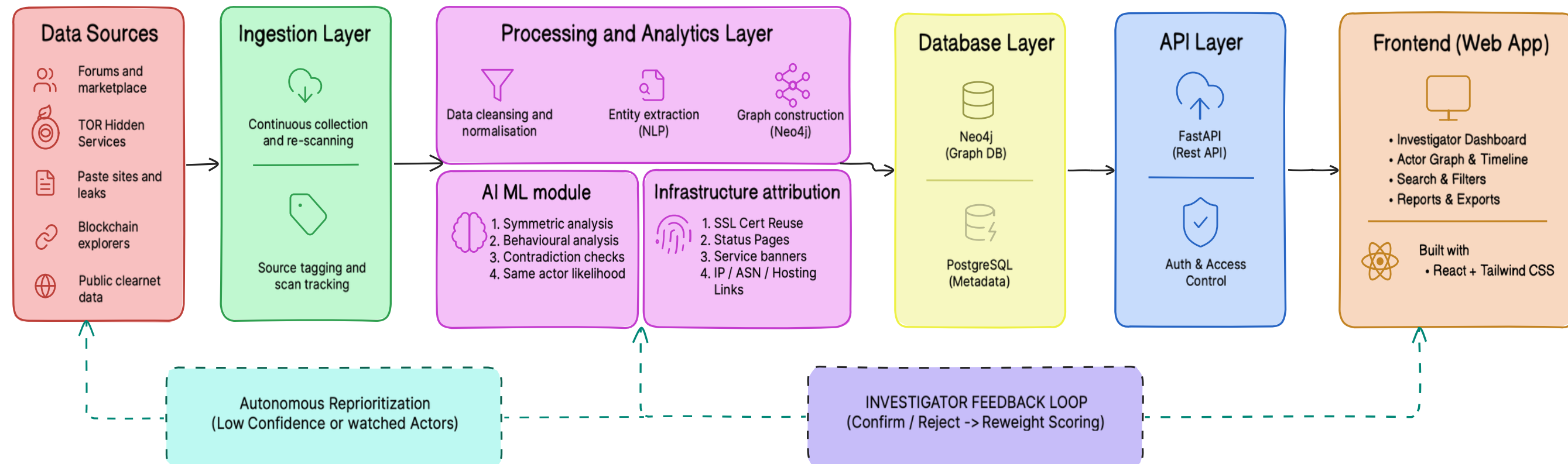


TECHNICAL APPROACH

METHODOLOGY



SYSTEM ARCHITECTURE



Analysis of Feasibility of Idea



Technical Feasibility

- Built using mature, open-source technologies (Neo4j, Python, NLP models, FastAPI, React) ensuring high feasibility.
- Modular architecture allows incremental development and integration.



Data Feasibility

- Synthetic and academic datasets + public sources are sufficient for model training and testing.
- Does not require access to live dark web or illegal marketplaces.



Operational Feasibility

- Automation reduces manual effort in collection, correlation and attribution.
- Investigator dashboard designed for real-world usability and workflow integration.



Legal & Ethical Feasibility

- System complies with legal, ethical and academic guidelines.
- No live scraping of illegal marketplaces; only ethical, self-created synthetic datasets used.



Economic Feasibility

- Uses open-source tools and cloud-agnostic design to minimize cost.
- Scalable deployment with efficient resource utilization.



Social Feasibility

- Addresses a critical need for law enforcement and cybersecurity agencies.
- Enhances public safety and trust through better attribution and disruption of threats.



Overall Feasibility: HIGH

The solution is technically sound, ethically responsible, operationally practical and socially impactful.

POTENTIAL CHALLENGES AND RISKS



Data Quality & Availability

- Public sources can be incomplete, noisy or inconsistent.
- Synthetic data may not capture real-world complexity fully.



Attribution Accuracy

- High risk of false positives due to anonymization, shared infrastructure and spoofing.
- Attribution confidence depends on quality of signals.



Evasion Techniques

- Threat actors continuously evolve (new anonymization tools, OPSEC tactics).
- Dynamic nature of dark web makes detection challenging.



Legal & Privacy Risks

- Mishandling data can lead to legal consequences.
- Ensuring compliance with privacy and data protection regulations is critical.



Model Limitations

- NLP and ML models may have bias or limited generalization.
- High computational cost for large-scale analysis.



Operational Risks

- System downtime, integration failures or data pipeline breaks can affect investigations.
- Dependence on third-party services/APIs.



Impact

These challenges can affect accuracy, reliability, legal compliance and overall trust in the system.

Strategies for Overcoming these Challenges



Improve Data Quality

- Multi-source data collection and cross-validation.
- Continuous data cleaning, deduplication and enrichment.



Increase Attribution Confidence

- Use multi-signal correlation (infrastructure, behavioral, financial, temporal).
- Apply confidence scoring and investigator-in-the-loop validation.



Stay Ahead of Evasion

- Continuously update infrastructure fingerprints and profiles.
- Monitor emerging TTPs and adapt detection logic.



Ensure Legal & Ethical Compliance

- Follow ethical data practices and maintain audit logs.
- Implement role-based access, data minimization and compliance frameworks.



Enhance Model Robustness

- Fine-tune models on diverse datasets and use ensemble approaches.
- Regularly evaluate, test and retrain models.



Strengthen Operational Resilience

- Build fault-tolerant pipelines with monitoring and alerts.
- Use containerized, scalable architecture with backups and failover mechanisms.



Outcome

Mitigated risks, higher accuracy, legal compliance and a resilient system trusted by investigators.

TARGET AUDIENCE



Law enforcement agencies:
Cyber crime units,
intelligence beureus



Faster investigations, actionable intelligence,
improved attribution accuracy and reduced
case resolution time.



Investigators & analysts:
OSINT analysts,
threat researchers



Streamlined workflows, better evidence
trails, higher confidence in findings and
reduced manual effort.



Judicial & legal system:
Prosecutors,
legal advisors



Stronger, well-attributed evidence
supporting prosecutions and improved
conviction outcomes.



**Policy makers &
government bodies:**
MHA , CERT-In , NIA



Data-driven insights for policy making,
threat monitoring and resource
allocation.



Public & digital community:
Citizens, Businesses,
Digital platforms



Safer digital spaces, reduced fraud
and cybercrime, increased trust in
online ecosystems.



Academic & research community:
Universities, Researchers,
Students



Access to ethical, labeled synthetic
datasets and tools for research
and innovation.

BENEFITS



SOCIAL BENEFITS

- Enhances public safety by identifying and disrupting malicious actors.
- Strengthens trust in law enforcement and digital platforms.
- Supports a safer and more accountable online environment for all.



ECONOMIC BENEFITS

- Reduces financial losses due to cybercrime, fraud and online scams.
- Optimizes investigative resources and reduces manual workload.
- Enables cost-effective, scalable investigations with automation and AI.



ENVIRONMENTAL BENEFITS

- Cloud-native and containerized architecture ensures efficient resource use.
- Reduced paper usage through digital, automated reporting and documentation.
- Promotes remote collaboration, lowering the carbon footprint of operations.



OPERATIONAL BENEFITS

- End-to-end workflow automation from data collection to actionable intelligence.
- Real-time insights with prioritization of high-risk actors and threats.
- Modular, scalable system adaptable to evolving threat landscapes.



KNOWLEDGE & INNOVATION BENEFITS

- Encourages research and innovation in cybersecurity and AI for social good.
- Provides an ethical sandbox with synthetic datasets for learning and experimentation.
- Builds domain expertise and capacity within the country.